Exploratory Data Analysis (EDA) Summary

1. Introduction

This report presents an exploratory data analysis (EDA) of Geldium Finance's customer dataset to support the development of an AI-powered delinquency prediction system. The analysis aims to assess data quality, identify missing or inconsistent information, detect potential risk factors, and examine relationships between customer financial characteristics, payment behavior, and delinquency outcomes. The insights gained from this analysis will provide a reliable foundation for data preprocessing, feature selection, and predictive modeling. Ultimately, the findings will support Geldium's Collections team in identifying at-risk customers and implementing more effective, data-driven intervention strategies while promoting responsible and explainable AI.

Objectives:

- Assess the overall quality and structure of the customer dataset.
- Identify missing values, inconsistencies, and potential anomalies.
- Detect patterns and key risk indicators associated with credit card delinquency.
- Recommend appropriate data preprocessing and missing-value treatment strategies.
- Prepare the dataset for AI-driven predictive modeling and risk assessment

2. Dataset Overview

The dataset contains 500 customer records with financial and behavioral variables relevant to delinquency prediction. Missing values are present in the Income, Loan_Balance, and Credit_Score variables, while Credit_Utilization includes values exceeding 100%, which require further validation to determine whether they represent valid observations or data quality issues. Key risk indicators include payment history, credit utilization, credit score, and debt-to-income ratio. After addressing missing values, encoding categorical variables, validating anomalous records, and applying appropriate preprocessing techniques, the dataset will be suitable for predictive modeling.

Key dataset attributes:

- Number of records (rows): 500 customers
- Number of attributes (columns): 19
- Target variable**:** Delinquent_Account (indicates whether a customer became delinquent)

Key variables:

Variable	**Description**
Customer_ID	Unique customer identifier
Age	Customer age
Income	Annual income
Credit_Score	Creditworthiness score
Credit_Utilization	Percentage of available credit used
Missed_Payments	Number of missed payments
Delinquent_Account	Target variable (0 = Not Delinquent, 1 = Delinquent)
Loan_Balance	Outstanding loan amount
Debt_to_Income_Ratio	Debt relative to income
Employment_Status	Employment category
Account_Tenure	Length of customer relationship
Credit_Card_Type	Card category
Location	Customer location
Month_1 – Month_6	Monthly payment history (On-time, Late, Missed)

Data types:

A) Categorical Variables

- Customer_ID
- Employment_Status

- Credit_Card_Type
- Location
- Month_1 to Month_6 payment status

B) Numerical Variables

- Age
- Income
- Credit_Score
- Credit_Utilization
- Missed_Payments
- Loan_Balance
- Debt_to_Income_Ratio
- Account_Tenure
- Delinquent_Account (binary target)

2.1 Notable Missing or Inconsistent Data

- The dataset contains missing values in several financial variables that require preprocessing before predictive modeling.
- Potential anomalies were observed in Credit_Utilization, where some values exceed 100% and should be validated.
- Categorical variables should be checked for consistent formatting before encoding.
- Customer identifiers should be verified to ensure record uniqueness.

2.2 Key Anomalies

- Further analysis is required to determine the presence of outliers in Income, Loan_Balance, and Debt_to_Income_Ratio.
- Credit Utilization values above 100% or below 0% would indicate data quality issues.
- Unusual Age values (e.g., below 18 or extremely high ages) may represent data entry errors.
- Unexpected payment history values outside "On-time," "Late," or "Missed" should be flagged for review.

2.3 Early Indicators of Delinquency Risk

- Higher numbers of Missed Payments are likely associated with increased delinquency risk.
- High Credit Utilization may indicate financial stress and potential repayment difficulties.
- Lower Credit Scores often suggest a higher likelihood of delinquency.
- High Debt-to-Income Ratios may indicate customers carrying excessive debt relative to their income.
- Repeated "Late" or "Missed" statuses in Month_1–Month_6 payment history could serve as strong warning signs of future delinquency.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

3.1 Notable Missing Data

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MISSING VALUES SUMMARY
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              Missing Values  Percentage (%)
Income                    39             7.8
Credit_Score               2             0.4
Loan_Balance              29             5.8

Critical Features Missing Values:
Income: 39 missing values
Loan_Balance: 29 missing values
Credit_Score: 2 missing values
Credit_Utilization: 0 missing values
Debt_to_Income_Ratio: 0 missing values
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Figure 3.1: Summary of Missing Values in the Datase

- Income contains 39 missing values (7.8%), making it the variable with the highest proportion of missing data. Since income is an important indicator of a customer's repayment capacity, these missing values should be addressed before predictive modeling.
- Loan_Balance contains 29 missing values (5.8%), which may affect the assessment of customers' debt burden and financial risk.

- Credit_Score contains 2 missing values (0.4%). Although the percentage is small, credit score is an important predictor of delinquency risk and should be imputed or otherwise handled.
- Credit_Utilization and Debt_to_Income_Ratio contain no missing values, indicating complete data for these important risk-related features.

3.2 Missing Data Treatment Strategy

Feature	Missing Values	Selected Method	Justification
Income	39 (7.8%)	Median Imputation	Income data may contain extreme values. Median imputation reduces the impact of outliers while preserving the overall distribution.
Loan_Balance	29 (5.8%)	Median Imputation	Loan balances can be highly skewed. Median imputation provides a robust estimate and retains valuable customer records.
Credit_Score	2 (0.4%)	Median Imputation	The number of missing values is very small, making median imputation a simple and effective solution.

Median imputation was selected because the missing values are relatively limited in number and occur within numerical financial variables that may contain outliers. This method preserves the dataset by retaining all customer records while minimizing the influence of extreme values. It also maintains the overall distribution of the data, making it suitable for predictive modeling. Although synthetic data generation was considered, it was not required because the level of missingness is low and can be effectively addressed using standard imputation techniques.

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

4.1 Correlation Analysis

A correlation analysis was conducted to examine the relationships between numerical variables and the target variable (Delinquent_Account). The results indicate that all numerical variables have very weak linear correlations with delinquency, suggesting that no single feature is a strong standalone predictor.

Correlations Observed

Variable	Correlation	Interpretation
Income	0.045	Very weak positive relationship with delinquency.
Credit_Score	0.035	Very weak positive relationship.
Debt_to_Income_Ratio	0.034	Slight positive relationship with delinquency.
Credit_Utilization	0.034	Slight positive relationship with delinquency.
Age	0.023	Negligible relationship.
Loan_Balance	-0.004	No meaningful relationship.
Missed_Payments	-0.026	Very weak negative relationship.
Account_Tenure	-0.040	Very weak negative relationship.

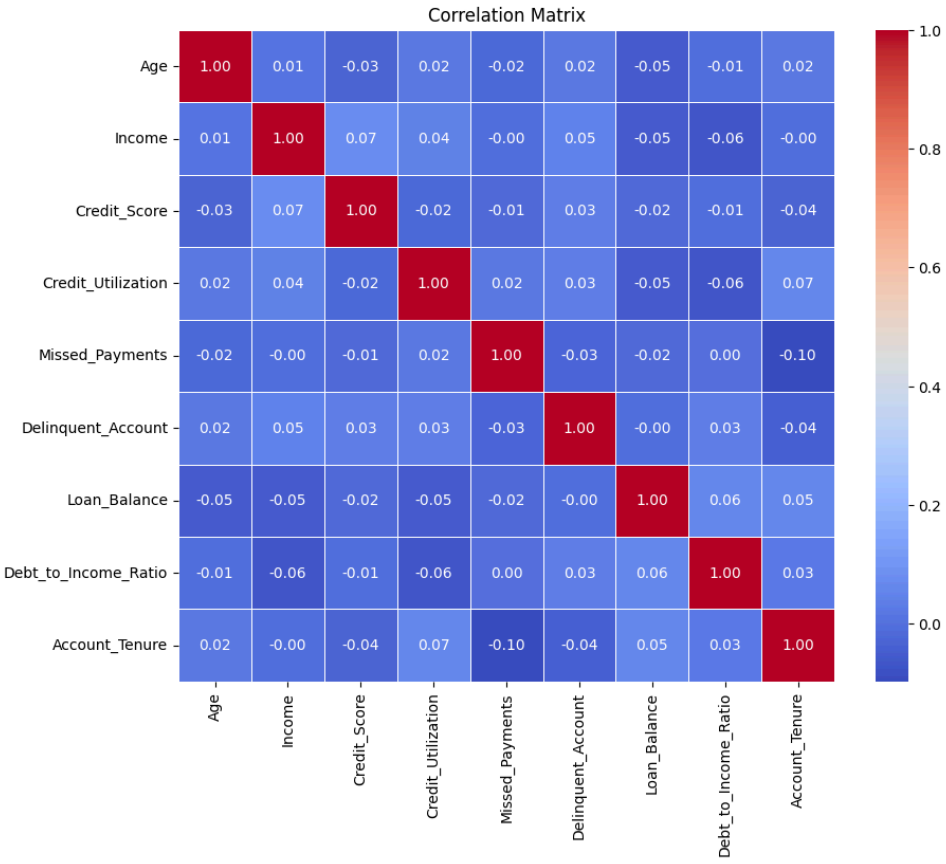

Figure 4.1: Correlation matrix showing the relationships among numerical variables in the dataset.

4.2 High-Risk Indicators

High-Risk Indicator	Explanation
Credit Utilization	Customers with higher credit utilization showed a slightly higher average delinquency rate, suggesting that extensive credit usage may indicate financial stress.
Debt-to-Income Ratio	A higher debt-to-income ratio may reduce a customer's repayment capacity and increase delinquency risk.
Account Tenure	Customers with shorter account tenure showed a slightly higher delinquency rate, suggesting newer customers may represent a higher-risk group.
Employment Status	Unemployed customers exhibited the highest delinquency rate among employment categories, indicating employment status may influence repayment behavior.
Payment History	Repeated late or missed payments may indicate repayment difficulties and help identify customers at risk of delinquency.

4.3 Key Insights

- Correlation analysis revealed weak linear relationships between numerical variables and delinquency.
- Credit Utilization and Debt-to-Income Ratio showed slightly higher values among delinquent customers.
- Employment Status exhibited small differences in delinquency rates, with unemployed customers showing the highest rate.
- Inconsistent employment labels (e.g., EMP, Employed, and employed) should be standardized before model training.
- Some Credit Utilization values exceeded 100%, requiring validation before predictive modeling.

5. AI & GenAI Usage

Generative AI tools (e.g., ChatGPT) were used to assist with exploratory data analysis (EDA), identify data quality issues, recommend preprocessing techniques, and summarize key insights. AI-generated suggestions were reviewed and validated against the dataset before being incorporated into the analysis.

Example AI prompts used:

- "Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency."
- "Identify the top variables most likely to predict delinquency and explain why they are important."
- "Suggest an imputation strategy for missing values in this dataset based on industry best practices."
- "Propose best-practice methods to handle missing credit utilization data for predictive modeling."
- "Analyze the relationship between customer attributes and delinquency outcomes and identify potential risk factors."
- "Interpret the correlation results and highlight any unexpected findings that may require further investigation."

6. Conclusion & Next Steps

The exploratory data analysis identified missing values, data quality issues, and potential risk factors relevant to delinquency prediction. Median imputation was selected to address missing numerical data, while anomalies and inconsistent categorical values were identified for further validation. Although the correlation analysis showed only weak linear relationships, variables such as Credit Utilization, Debt-to-Income Ratio, Payment History, and Employment Status remain important for predictive modeling. The next phase of the project will focus on data preprocessing, feature engineering, model development, and evaluating the predictive performance of AI-based delinquency risk models.